

ModiDeC: a multi-RNA modification classifier for direct nanopore sequencing

Supplementary Information

Table of Contents

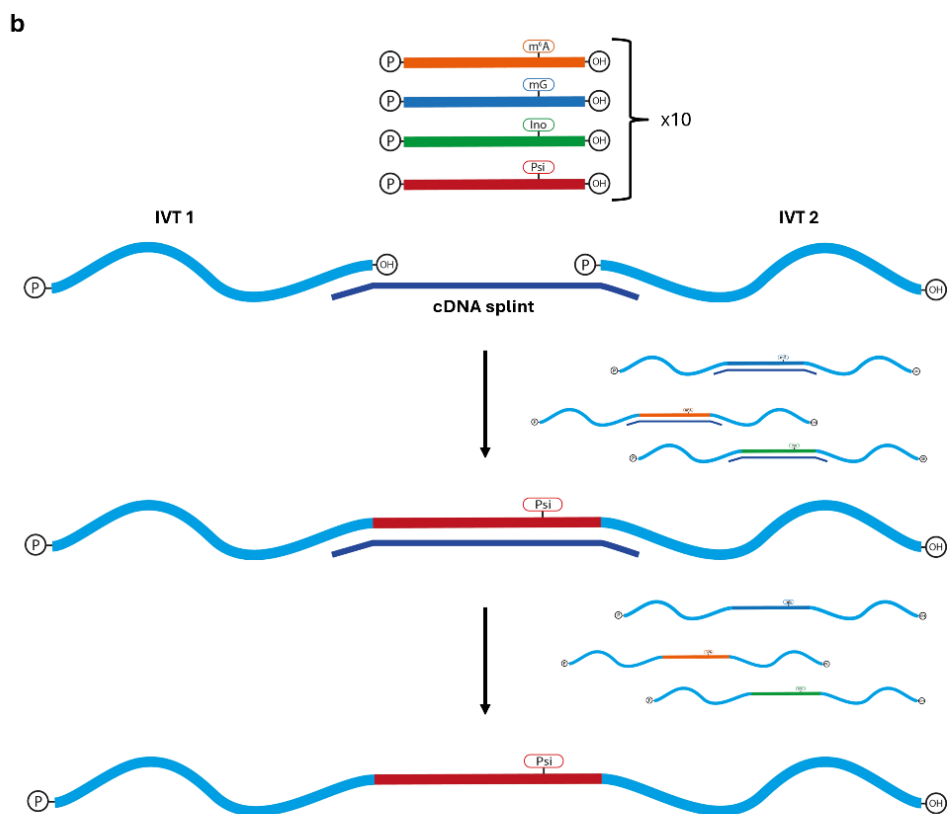
SI 1. Creation of training data sets for ModiDeC	3
Supplementary Table 1	3
Supplementary Fig. 1	6
SI 2. Testing neural network efficiency over reads used for the training phase.....	7
Supplementary Fig. 2	9
.....	10
Supplementary Table 2	10
SI 3. ModiDeC validation results on unmodified read region in 004 ONT kit.....	11
Supplementary Fig. 3	11
SI 4. ModiDeC validation results for SQK-RNA002	12
Supplementary Fig. 4	12
Supplementary Fig. 5	13
Supplementary Table 3	14
SI 5. Analysis of HEK293T cells with Dorado: selective organelle and ribosomal RNA.....	15
Supplementary Fig. 6	Error! Bookmark not defined.
Supplementary Fig. 7:	16
Supplementary Table 4	17
SI 6. ModiDeC analysis on human blood sample	18
Supplementary Fig. 8.	18
Supplementary Fig. 9	19
Supplementary Fig. 10	20
Supplementary Fig. 11	21
SI 7. ModiDeC RNA modification features extraction	22
Supplementary Table 5	22
SI 8. ModiDeC user graphic interface	23
SI 9. False positive analysis on IVT	24
Supplementary Fig. 12	25
SI 10. ModiDeC architecture.....	25
Supplementary Fig. 13	26

SI 1. Creation of training data sets for ModiDeC

Supplementary Table 1: List of all oligonucleotides used for training ModiDeC. The incorporated modified bases **pseudouridine (Ψ)**, **2'-O-methylguanosine (Gm)**, **N⁶-methyladenosine (m⁶A)**, **inosine (I)**, **N¹-methyladenosine (m¹A)** and **2'-O-methyladenosine (mA)** are highlighted and color-coded as they appear in the main text and supplementary section.

Name	Full sequence	Internal modification
unmod.	GAUACGGGAGACAGCCACCGAAUACGGGAGACAGCCACCUC	none
Psi #1	GAUACGGGAGUCAGCCACCGAAUACGGGAG Ψ CAGCCACCUC	Pseudouridine (Ψ)
Psi #2	GAGCCAUGAUUAAGAGGGAGGAAGCCAUGAU Ψ AAGAGGGAUC	Pseudouridine (Ψ)
Psi #3	GAGGUGACUCUAGAUAAACCGGAAGGUGACUC Ψ AGAUAAACCUC	Pseudouridine (Ψ)
Psi #4	GAACCACGGGUGACGGGGAGGAAACCACGGG Ψ GACGGGGAUC	Pseudouridine (Ψ)
Psi #5	GAUGCGUGCAUUUAUCAGAGGAUUGCGUGCA Ψ UUUAUCAGAUC	Pseudouridine (Ψ)
Psi #6	GAAGGCCAAGUAGCCCGUGGGAAAGGCCAAG Ψ AGCCCGUGUC	Pseudouridine (Ψ)
Psi #7	GAAUGCCACCUAGGGCAAGGGAAAUGCCACC Ψ AGGGCAAGUC	Pseudouridine (Ψ)
Psi #8	GAAUUGAUUCUAGCAGGGGGAAAUGAUUC Ψ UAGCAGGGUC	Pseudouridine (Ψ)
Psi #9	GACCUAGGGAUUCUCCGAAGGAACCUAGGGA Ψ CUUCCGAAUC	Pseudouridine (Ψ)
Psi #10	GAUCGAGGUCUCUGCAACCGGAUUCGAGGUC Ψ CUGCAACCUC	Pseudouridine (Ψ)
Gm #1	GAUACGGGAGGCAGCCACCGAAUACGGGAGGmCAGCCACCUC	2'-O-Methylguanosine (Gm)
Gm #2	GACCGGGGAGGUAGUGACGGGAACCGGGGAGGmUAGUGACGUC	2'-O-Methylguanosine (Gm)
Gm #3	GAUAAUAACGGGUCUGUGAGGAUAAUAACGmGUCUGUGAUC	2'-O-Methylguanosine (Gm)
Gm #4	GACGUAGUUGGAUCUUGGGGAACGUAGUUGGmAUUUGGGUC	2'-O-Methylguanosine (Gm)
Gm #5	GAACUUCUAGAGGGACAAGGAAACUUCUAGmAGGGACAAUC	2'-O-Methylguanosine (Gm)
Gm #6	GAUGUGAAUUGCAGGACACGGAUUGGAAUUGmCAGGACACUC	2'-O-Methylguanosine (Gm)
Gm #7	GACAUUGGAGGGCAAGUCUGGAACAUUGGAGGmGCAAGUCUUC	2'-O-Methylguanosine (Gm)
Gm #8	GACUUGAGCAUGCGAGAGGUGGAACUUGCAUUGmCGAGAGGUUC	2'-O-Methylguanosine (Gm)
Gm #9	GAAAUUUUCGCCUUUUACGGAAAACUUCUUGmCCUUUUACUC	2'-O-Methylguanosine (Gm)
Gm #10	GAGGACUAGAGGCCGUUUGGGAAGGACUAGAGmGCCGUUUGUC	2'-O-Methylguanosine (Gm)
m ⁶ A #1	GAUACGGGAGACAGCCACCGAAUACGGGAGm ⁶ ACAGCCACCUC	N ⁶ -Methyladenosine (m ⁶ A)
m ⁶ A #2	GAGUGCCAGGACCGACCAUGGAAGUGCCAGGm ⁶ ACCGACCAUUC	N ⁶ -Methyladenosine (m ⁶ A)
m ⁶ A #3	GACACCAGUGACUCCAUAGGAACACAGUGm ⁶ ACUCCCAUAUC	N ⁶ -Methyladenosine (m ⁶ A)
m ⁶ A #4	GAUAAACGAGACCGUCUAGGGAAUAAACGAGm ⁶ ACCGUCUAGUC	N ⁶ -Methyladenosine (m ⁶ A)
m ⁶ A #5	GAGUUAAGAGACUGAAUCUGGAAGUUAAGAGm ⁶ ACUGAAUCUUC	N ⁶ -Methyladenosine (m ⁶ A)
m ⁶ A #6	GAAGUACAAAACAAUCAUUGGAAAGUACAAAACm ⁶ ACAAUCAUUC	N ⁶ -Methyladenosine (m ⁶ A)
m ⁶ A #7	GAUUCAGAGAACCACUUGAGGAAUUCAGAGAm ⁶ ACCACUUGAUC	N ⁶ -Methyladenosine (m ⁶ A)
m ⁶ A #8	GAUAAACUUAACUCCAAAAGGAAUAAACUUAACm ⁶ ACUCCAAAUC	N ⁶ -Methyladenosine (m ⁶ A)
m ⁶ A #9	GAUCACUCGAACUUAAGCGGAUUCACUCGAm ⁶ ACUUCAAGCUC	N ⁶ -Methyladenosine (m ⁶ A)
m ⁶ A #10	GAUUGUGGUAACGUCCCCAGGAUUGUGGUAm ⁶ ACGUCCCCAUC	N ⁶ -Methyladenosine (m ⁶ A)

m ⁶ A #11	AACGCCUGGCm ⁶ AGCCGGAAGCC	N ⁶ -Methyladenosine (m ⁶ A)
Inosine #1	GAUACGGGAGACAGCCACCGGAAUACGGGAGICAGCCACCUC	Inosine (I)
Inosine #2	GAGCUCGCUUAGCAUGCGAGGAAGCUCGCUUIGCAUGCGAUC	Inosine (I)
Inosine #3	GAGAGGGUCGAGAUCCACUGGAAGAGGGUCGICAUCCACUUC	Inosine (I)
Inosine #4	GAUUGCGGAUACCGAACGAGGAUUUGCGGAUICCGAACGAUC	Inosine (I)
Inosine #5	GAGAGCUGGCACGAGGCUCGGAAGAGCUGGCICGAGGCUCUC	Inosine (I)
Inosine #6	GAGCCCCAAAACGGCCGAUGGAAGCCCCAAACGGCCGAUUC	Inosine (I)
Inosine #7	GAUGGAGAGCACGAGGCCCGGAUUGGAGAGCICGAGGCCUC	Inosine (I)
Inosine #8	GAAUCUUGUCAUUGUCGUCGGAUUCUUGUCIAUGUCGUCUC	Inosine (I)
Inosine #9	GAUUAUGUCAGGCACCCUGGAUUAUACGUCIGGCACCCUUC	Inosine (I)
Inosine #10	GAGGAACGGCACAGUUAUUGGAAGGAACGGCICAGUUAUUUC	Inosine (I)
m ¹ A #1	CCACm ¹ AAACCAUGGUGAGCAA	N ¹ -Methyladenosine (m ¹ A)
m ¹ A #2	AACGCCUGGCm ¹ AGCCGGAAGCC	N ¹ -Methyladenosine (m ¹ A)
m ¹ A #3	AACGCCUGGCAGCCGGAAGCC	none
mA #1	AACGCCUGGCmAAGCCGGAAGCC	2'-O-methyladenosine (mA)

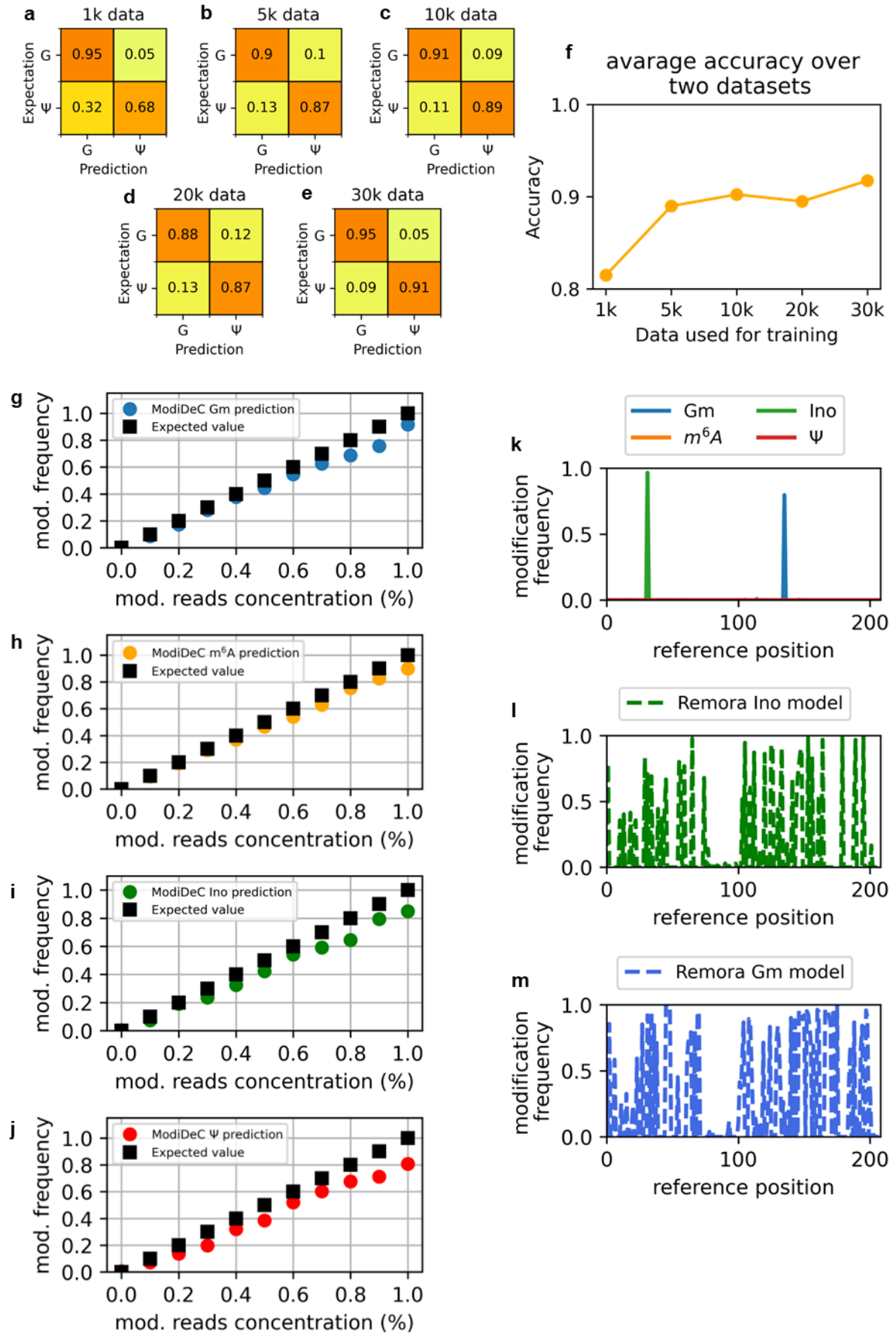


Supplementary Fig. 1: Oligonucleotide design and incorporation of oligonucleotides into sequencing construct via splinted ligation. (a) Oligonucleotides contain the modified and unmodified nucleotide (red) flanked by a 9-mer sequence of choice (bold, black). (b) Oligonucleotides are incorporated into a larger RNA construct via splinted ligation. A cDNA splint complementary to the 5' and 3'- ends of the IVT RNA and the oligonucleotides aligns the components for ligation by T4 RNA Ligase 2. Following ligation, the cDNA splint is digested by DNase, leaving behind the ligated product. (c) Ligation of sequencing constructs containing multiple oligonucleotides. Two complementary cDNA splints are supplied, one for each oligonucleotide. The alignment and DNase digestion process is as described in panel (b).

SI 2. Testing neural network efficiency over reads used for the training phase

In this part of the study we investigated whether we could identify a certain number of reads to use for the training phase. To this end, we trained ModiDeC to analyze pseudouridine RNA modifications (Ψ) using different amounts of training data. Specifically, we used 1,000, 5,000, 10,000, 20,000, and 30,000 reads for the training phase and then tested the efficiency of the neural network on new constructs (different from the training constructs) that contain a Ψ modification at a specific position. An unmodified motif is also present in the testing construct and used for the analysis. The confusion matrices for 1,000, 5,000, 10,000, 20,000, and 30,000 reads (Supplementary Fig. 2a–e) show prediction versus expectation for the modified position (Ψ) and for the unmodified position (G). We observed that increasing the number of reads for the training increased the percentage of correctly called Ψ (low right squares on the heatmaps), reaching a maximum value for 30,000 reads (Supplementary Fig. 2e). The unmodified position seems to follow a different trend than the modified one. To better understand the global effect of data training of neural network performance, we averaged the accuracy values of the main diagonal for each heatmap and then plotted the results versus the number of reads (Supplementary Fig. 2f). This plot shows that general performance depends on the number of training data used. We find the curve somewhat saturated at 10,000 reads, which suggests using a certain minimum number of reads in the training phase. However, it is difficult to reach a general conclusion because the results presented in Fig. 1e and Supplementary Fig. 4 show different frequencies of modification detection that depend on the RNA motif, although uniform training data were used during the training phase (each motif was trained with approximately 30,000 reads). Despite these mixed results, we suggest using minimum 10,000 reads as a reference value for future training.

In this section, we also investigated the limit of detection (LOD) and quantification (LOQ) of ModiDeC by virtual titration analysis (See Supplementary fig. 2g-j). It consists on generating test batches with selected modification frequency by mixing modified and unmodified reads. We generated eleven titration batches (from 0% to 100%, with variation of 10%, expected modification frequency) for each of the four modification types and let ModiDeC analyze the data to see if the predicted and expected values matched. For each modification type titration batch, modified and



Supplementary Fig. 2: Confusion matrices on test data using ModiDeC trained with 1,000 (a), 5,000 (b), 10,000 (c), 20,000 (d) and 30,000 (e) reads. Decimal fraction values are written inside each square. It is possible to observe an increase of true positives for pseudouridine predictions by increasing the training data batch. f) True-positive average values (y axis) plotted against the size of the training data batch (x axis). True positive average values were calculated by averaging the main diagonal of each confusion matrix. Performance seems to saturate after 10,000 training data. Virtual titration experiments for Gm (g), m⁶A (h), Ino (i), and Ψ (j). eleven titration experiments were generated (with different modification frequency) for each modification type. Comparison between ModiDeC (k) and Remora Gm (l) / Ino (m) models. Remora shows several false positives compared to ModiDeC.

unmodified reads were randomly selected to generate a batch size of 1000 reads from the test data (see Fig. 1e for test data). The LOD and LOQ for each modification type were calculated using the following formula:

$$LOD = 3.3 * \left(\frac{\sigma}{S}\right) \quad (\text{SI eq. 1})$$

$$LOQ = 10 * \left(\frac{\sigma}{S}\right) \quad (\text{SI eq. 2})$$

Where σ is the standard deviation of the data and S is the linear regression slope obtained by fitting the virtual titration experiment. The analysis shows similar LOD and LOQ for all the four modification types. The average values are $LOD_{avg.} = 1.1\%$ and $LOQ_{avg.} = 3.2\%$.

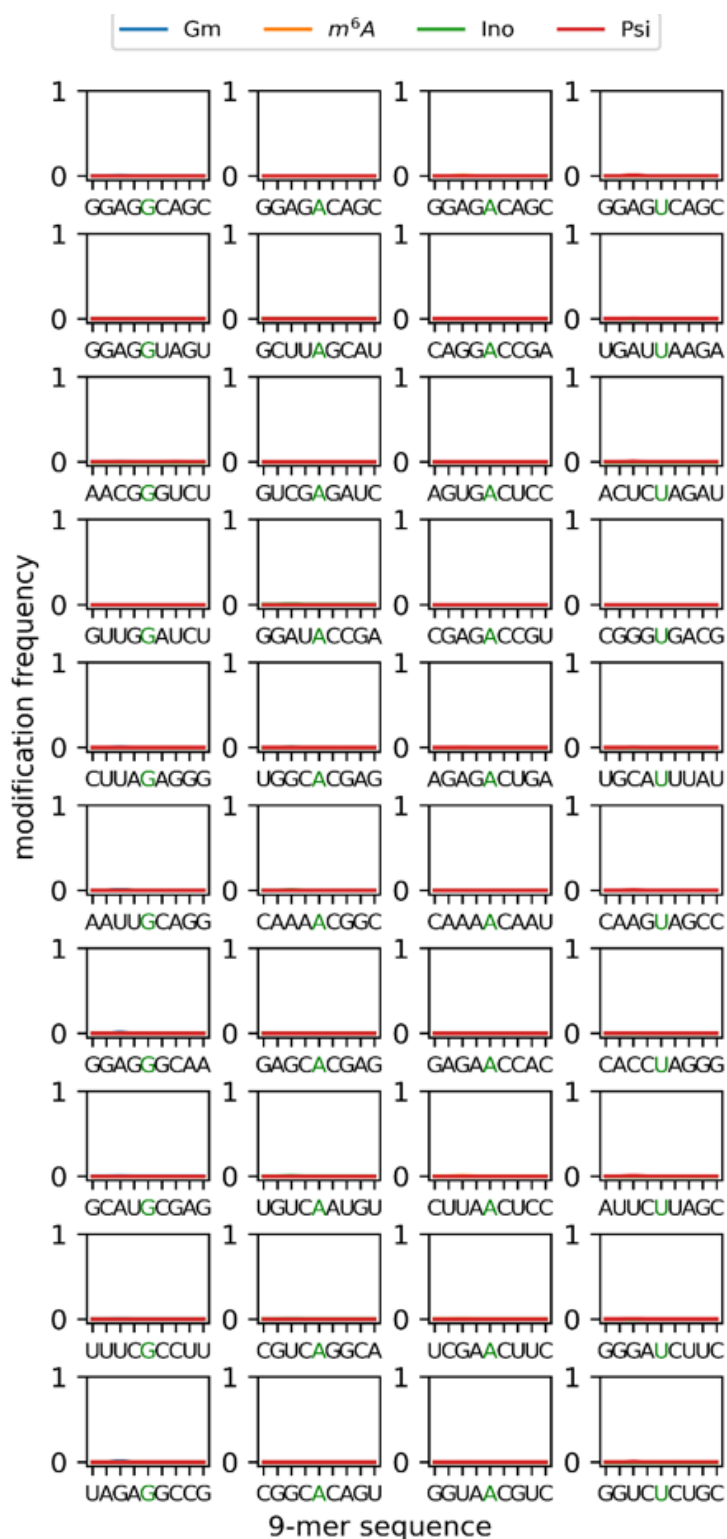
Next, we compared ModiDeC and Remora model. In detail, we trained Remora two times to detect Gm (Remora Gm model) and Ino (Remora Ino model) using the ModiDeC training data. For each modification type, 30k chunks modified and 30k chunks unmodified were used for each motif (see Fig. 1 main text for the motifs). In total, Remora was trained with 600k chunks for each modification type for 20 epochs. The comparison between the two models shows that ModiDeC can detect the two expected positions, without any clear false positive, while Remora presents several false positive over the construct.

Supplementary Table 2: Modified 9-mer sequences for the 40 oligos analyzed in Fig. 1e measured with the 004 kit. The table shows the true-positive, false-negative and modification misscall percentages obtained using ModiDeC. The expected true-positive percentage is 100% for all the 9-mers.

Sequence (Gm modified)	True positive	false negative	Modific. misscall	Sequence (Ino modified)	True positive	false negative	Modific. misscall
GGAG(Gm)CAGC	52%	48%	0%	GGAG(Ino)CAGC	94%	5%	1%
GGAG(Gm)UAGU	85%	15%	0%	GCUU(Ino)GCAU	92%	8%	0%
AACG(Gm)GUCU	92%	8%	0%	GUCG(Ino)GAUC	76%	24%	0%
GUUG(Gm)AUCU	97%	3%	0%	GGAU(Ino)CCGA	47%	53%	0%
CUUA(Gm)AGGG	94%	6%	0%	UGGC(Ino)CGAG	90%	10%	0%
AAUU(Gm)CAGG	81%	19%	0%	CAAA(Ino)CGGC	53%	47%	0%
GGAG(Gm)GCAA	78%	22%	0%	GAGC(Ino)CGAG	85%	15%	0%
GCAU(Gm)CGAG	96%	4%	0%	UGUC(Ino)AUGU	87%	13%	0%
UUUC(Gm)CCUU	92%	8%	0%	CGUC(Ino)GGCA	78%	22%	0%
UAGA(Gm)GCCG	70%	30%	0%	CGGC(Ino)CAGU	88%	12%	0%
Sequence (m ⁶ A modified)	True positive	false negative	Modific. misscall	Sequence (Ψ modified)	True positive	false negative	Modific. misscall
GGAG(m ⁶ A)CAGC	91%	9%	0%	GGAG(Ψ)CAGC	86%	14%	0%
CAGG(m ⁶ A)CCGA	85%	15%	0%	UGAU(Ψ)AAGA	52%	48%	0%
AGUG(m ⁶ A)CUCC	89%	11%	0%	ACUC(Ψ)AGAU	91%	9%	0%
CGAG(m ⁶ A)CCGU	86%	14%	0%	CGGG(Ψ)GACG	39%	51%	0%
AGAG(m ⁶ A)CUGA	91%	9%	0%	UGCA(Ψ)UUUAU	86%	14%	0%
CAAA(m ⁶ A)CAAU	90%	10%	0%	CAAG(Ψ)AGCC	63%	37%	0%
GAGA(m ⁶ A)CCAC	90%	10%	0%	CACC(Ψ)AGGG	82%	18%	0%
CUUA(m ⁶ A)CUCC	88%	12%	0%	AUUC(Ψ)UAGC	91%	9%	0%
UCGA(m ⁶ A)CUUC	81%	19%	0%	GGGA(Ψ)CUUC	86%	14%	0%
GGUA(m ⁶ A)CGUC	81%	19%	0%	GGUC(Ψ)CUGC	68%	32%	0%

SI 3. ModiDeC validation results on unmodified read region in 004 ONT kit

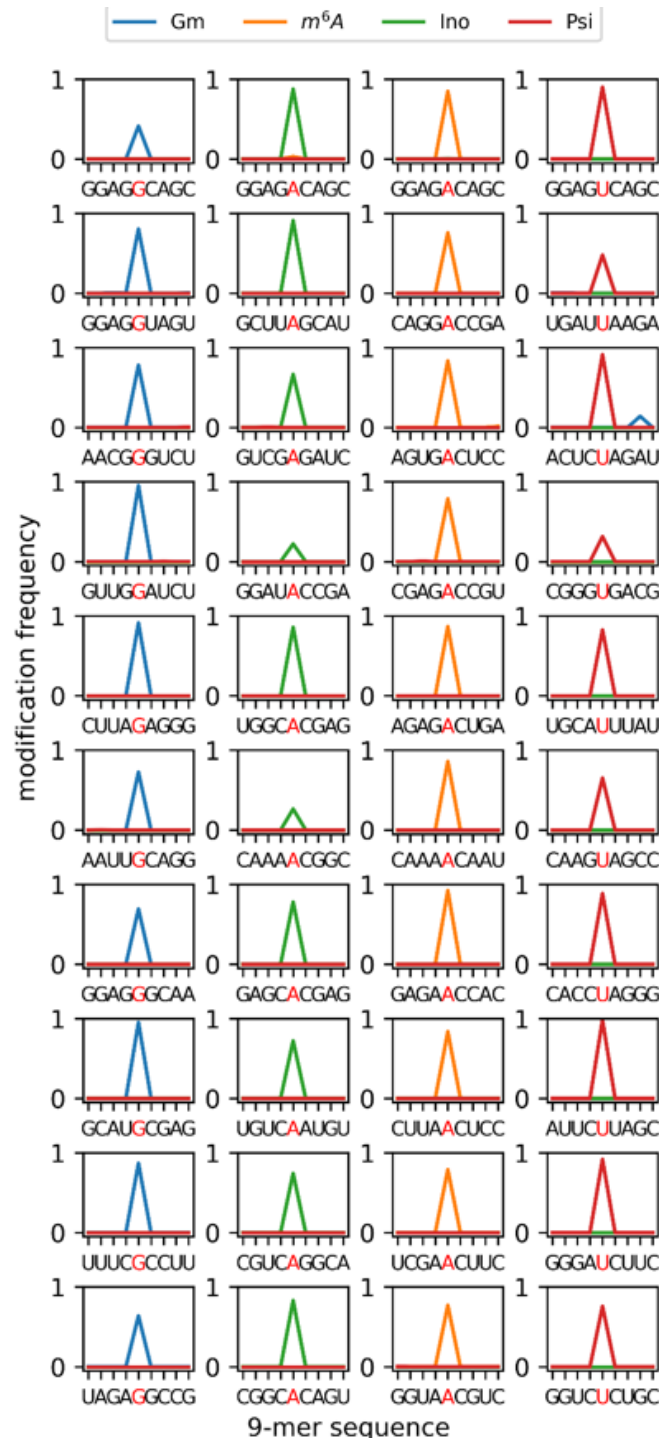
ModiDeC results for the SQK-RNA004 data for unmodified reads region. The sequence region is the same for modified (Fig. 1e) and unmodified (Supplementary Fig. 3) data.



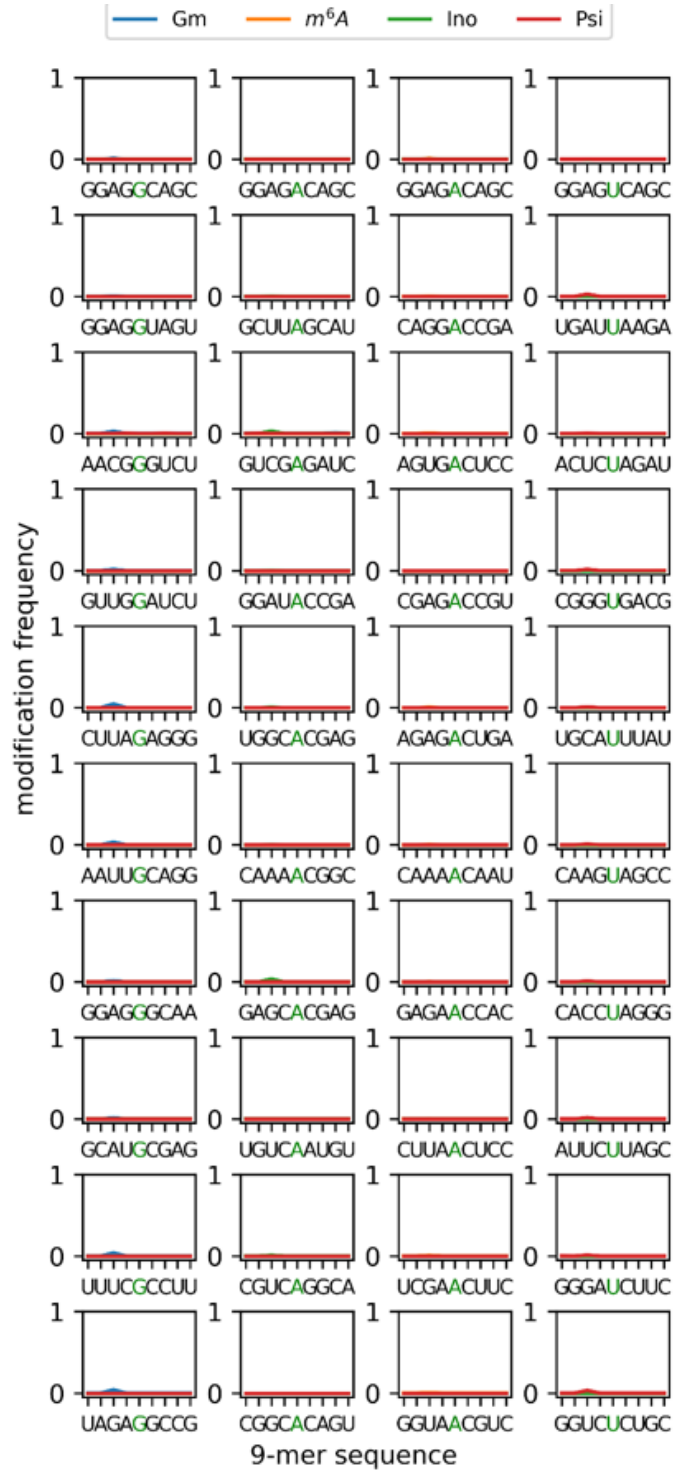
Supplementary Fig. 3: The modification frequency on the test data for the 40 motifs used in this work for unmodified reads for (SQK-RNA004 kit). The green letter shows the canonical letter that is modified in Fig. 1e. The unmodified region shows baseline results as expected.

SI 4. ModiDeC validation results for SQK-RNA002

Here we show the results obtained from the validation of ModiDeC after training the neural network on the SQK-RNA002 Kit. A comparison of the data in Fig. 1e and Supplementary Fig. 4 shows the frequency of detecting modifications is similar for the 002 and 004 kits. Supplementary Fig. 5 shows the results for the unmodified region. As expected, this region gave a practically baseline result, indicating that reads were not modified in the designed area.



Supplementary Fig. 4: The modification frequency on the test data for the 40 motifs used in this work (SQK-RNA002 kit). The red letter shows the modified base. We obtained similar results with the SQK-RNA004 kit data (see Fig. 1e).



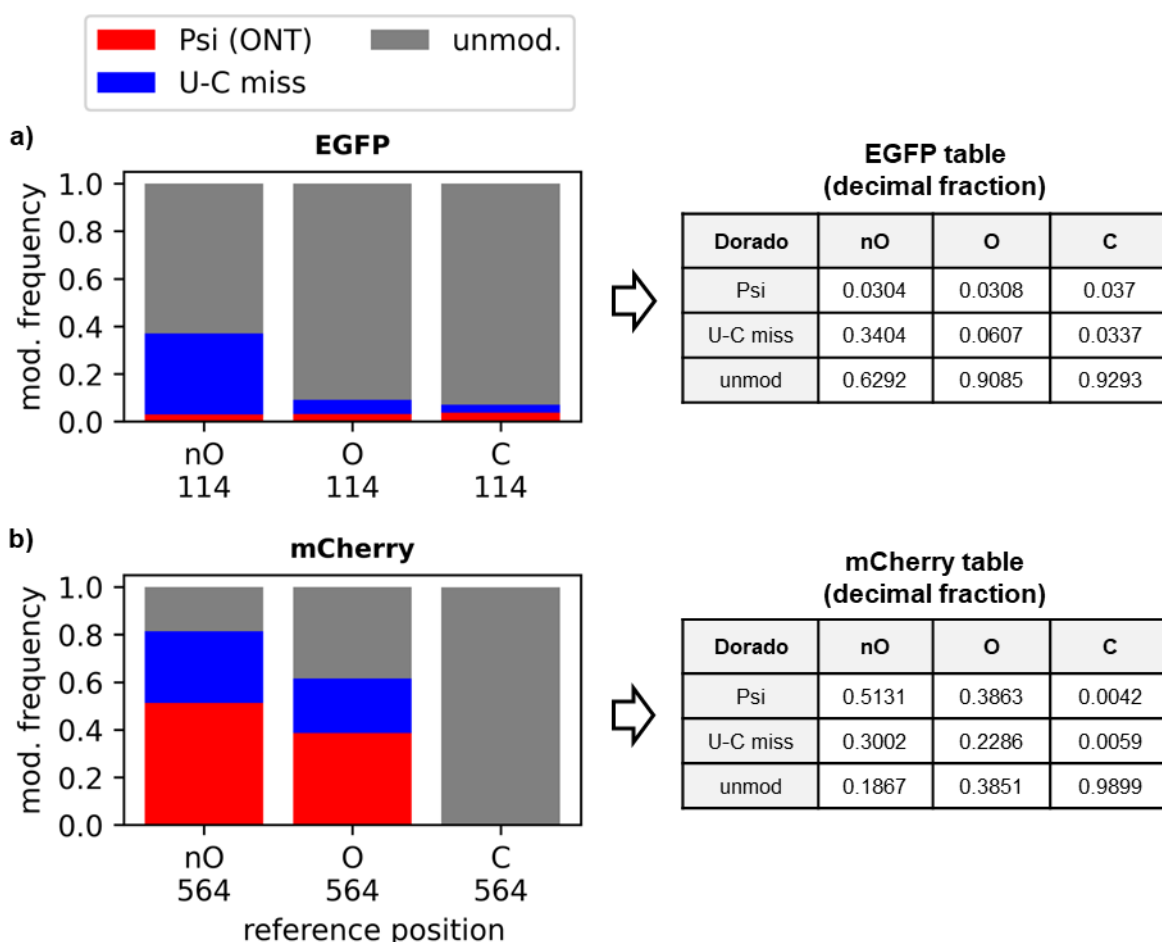
Supplementary Fig. 5: The modification frequency on the test data for the 40 motifs used in this work for unmodified reads with the SQK-RNA002 kit. The green letter shows the canonical letter that is modified in Fig. 1e. The unmodified region shows baseline results as expected.

Supplementary Table 3: Modified 9-mer sequence for the 40 oligos analyzed in Fig. 1e measured with the SQK-RNA002 kit. The table shows the true positive, false negative and modification miscalls obtained using ModiDeC for the oligos analysis. Results are shown for each modified 9-mer. The expected true positive value is 100% for all the 9-mers.

Sequence (Gm modified)	True positive	false negative	Modific. misscall	Sequence (Ino modified)	True positive	false negative	Modific. misscall
GGAG(Gm)CAGC	42%	58%	0%	GGAG(Ino)CAGC	88%	9%	3%
GGAG(Gm)UAGU	80%	20%	0%	GCUU(Ino)GCAU	91%	9%	0%
AACG(Gm)GUCU	78%	22%	0%	GUCG(Ino)GAUC	67%	23%	0%
GUUG(Gm)AUCU	95%	5%	0%	GGAU(Ino)CCGA	22%	78%	0%
CUUA(Gm)AGGG	91%	9%	0%	UGGC(Ino)CGAG	86%	14%	0%
AAUU(Gm)CAGG	73%	27%	0%	CAAA(Ino)CGGC	27%	73%	0%
GGAG(Gm)GCAA	70%	30%	0%	GAGC(Ino)CGAG	78%	22%	3%
GCAU(Gm)CGAG	95%	5%	0%	UGUC(Ino)AUGU	72%	28%	0%
UUUC(Gm)CCUU	97%	3%	0%	CGUC(Ino)GGCA	74%	26%	0%
UAGA(Gm)GCCG	64%	36%	0%	CGGC(Ino)CAGU	83%	17%	0%
Sequence (m ⁶ A modified)	True positive	false negative	Modific. misscall	Sequence (Ψ modified)	True positive	false negative	Modific. misscall
GGAG(m ⁶ A)CAGC	84%	25%	1%	GGAG(Ψ)CAGC	90%	10%	0%
CAGG(m ⁶ A)CCGA	76%	24%	0%	UGAU(Ψ)AAGA	48%	52%	0%
AGUG(m ⁶ A)CUCC	84%	16%	0%	ACUC(Ψ)AGAU	91%	9%	0%
CGAG(m ⁶ A)CCGU	79%	21%	0%	CGGG(Ψ)GACG	32%	68%	0%
AGAG(m ⁶ A)CUGA	87%	13%	0%	UGCA(Ψ)UUAU	83%	17%	0%
CAAA(m ⁶ A)CAAU	86%	14%	0%	CAAG(Ψ)AGCC	65%	35%	0%
GAGA(m ⁶ A)CCAC	92%	8%	0%	CACC(Ψ)AGGG	89%	11%	0%
CUUA(m ⁶ A)CUCC	84%	16%	0%	AUUC(Ψ)UAGC	98%	2%	0%
UCGA(m ⁶ A)CUUC	79%	21%	0%	GGGA(Ψ)CUUC	92%	8%	0%
GGUA(m ⁶ A)CGUC	77%	23%	0%	GGUC(Ψ)CUGC	76%	24%	0%

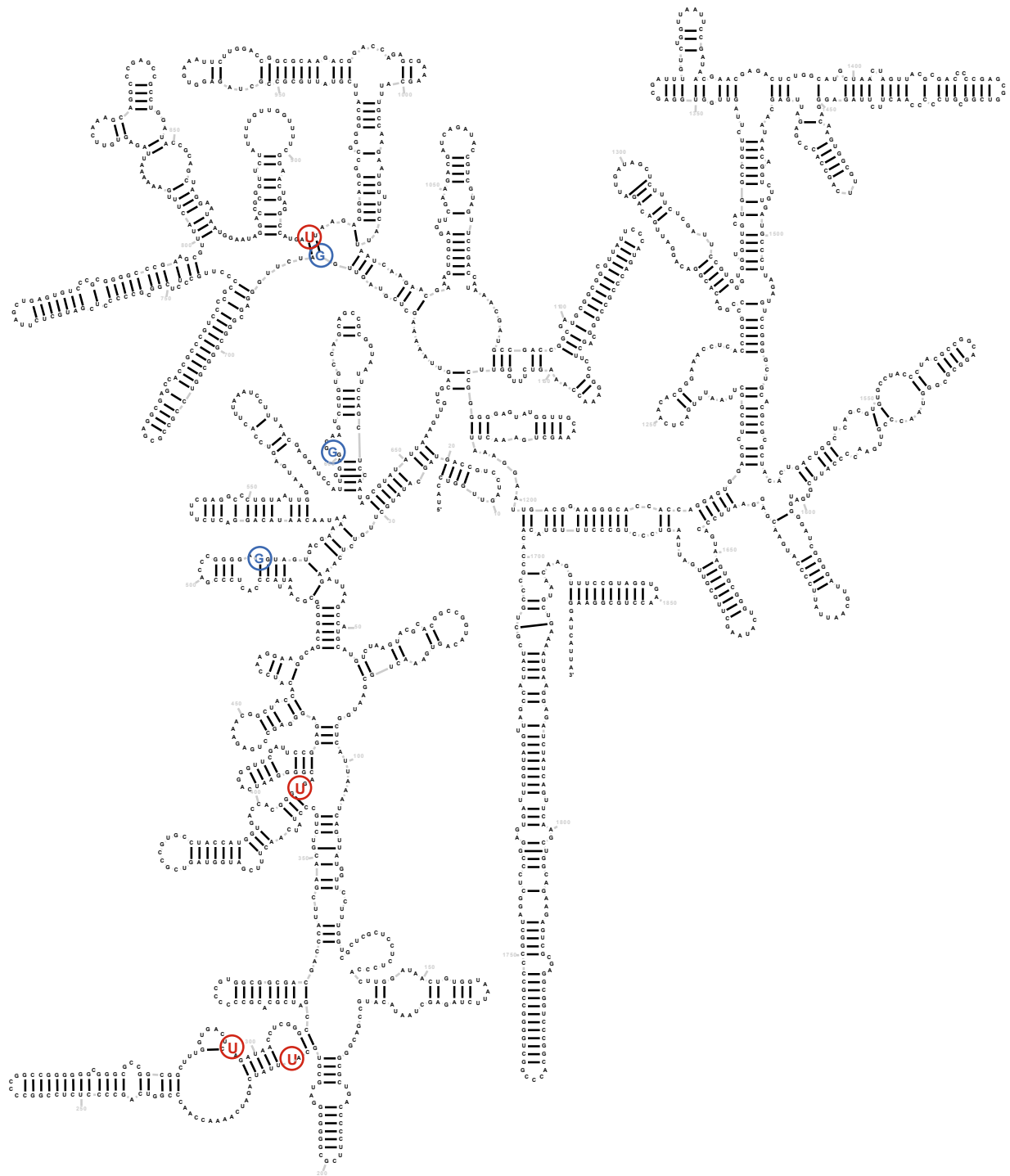
SI 5. Analysis of HEK293T cells with Dorado: selective organelle and ribosomal RNA

We performed Dorado analysis on HEK293T cells containing orthogonally translating film-like organelles. In details, EGFP and mCherry fluorescence protein sequences were analyzed using Dorado to quantify the amount of pseudouridine in specific sequence positions. Supplementary Fig. 6 shows the modification frequencies of pseudouridine (Psi), U–C mismatch, and unmodified reads detected by the Dorado model (Oxford Nanopore Technology) for the following three systems: without organelle (nO), with organelle (O), and control (C). The expected reference position of modification is shown below each bar plot (114 for EGFP and 564 for mCherry). Tables containing the results of the Dorado analysis for the three samples are shown next to each bar plot in Supplementary Fig. 6.



Supplementary Fig. 6: Dorado analysis on EGFP (a) and mCherry (b) sequences for systems without organelle (nO), with organelle (O), and control (C). On the right of the plot, the Dorado analysis values for the three samples are tabulated.

Additional information about the ModiDeC analysis on ribosomal RNA are reported below in the Supplementary Table 4.



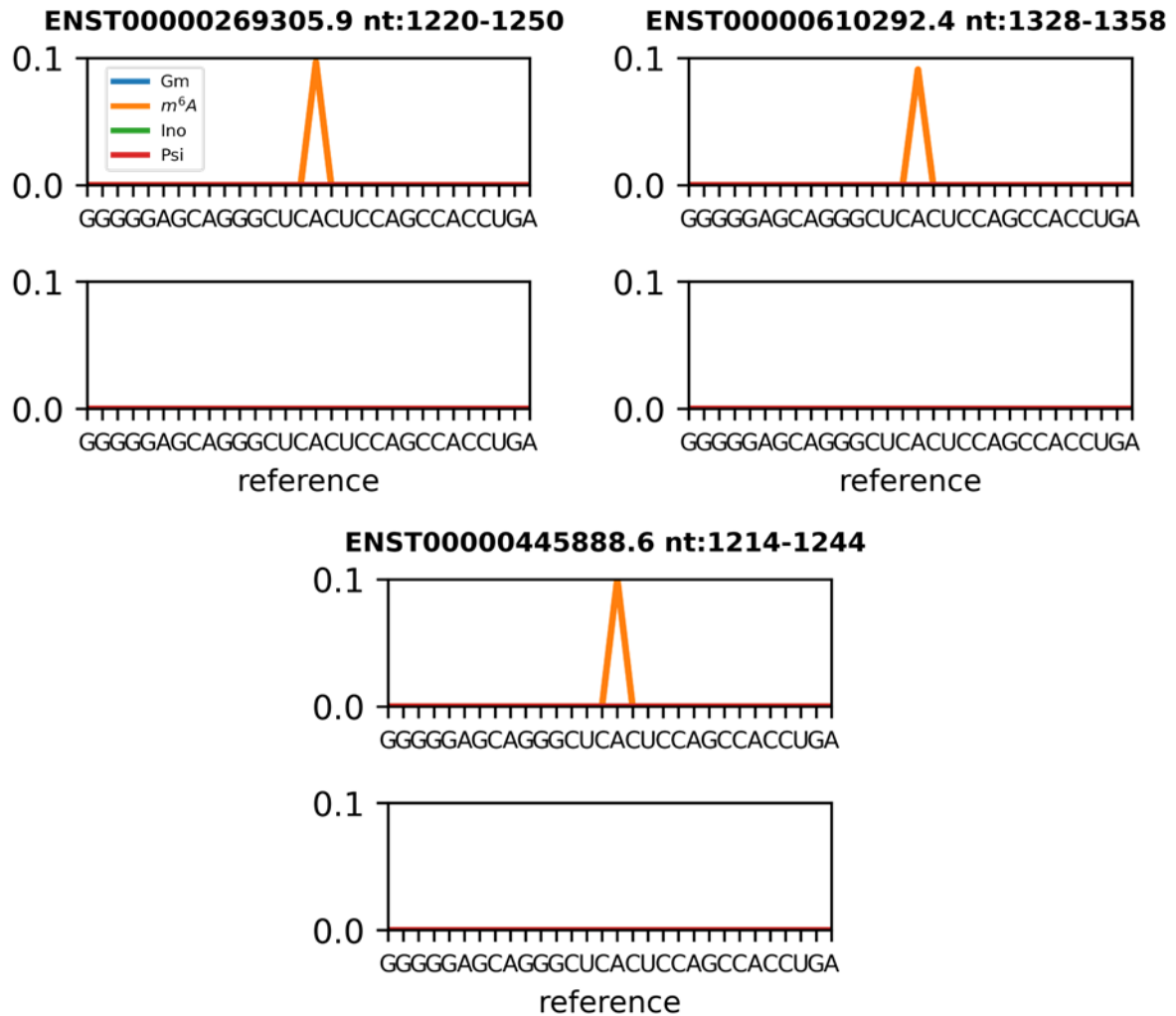
Supplementary Fig. 7: RNA45SN1 RNA complete structure. The letters highlighted are the modified bases with trainings-motives as recognized by ModiDeC (red = Psi, blue = Gm). The numbers next to these letters indicate the reference position.

Expected 18S position + mod. frequency	Dorado (Ψ) Position + mod. frequency (sample 1)	Dorado (Ψ) Position + mod. frequency (sample 2)	ModiDeC Position + mod. frequency (sample 1)	ModiDeC Position + mod. frequency (sample 2)
217-218 (100%)	217 (64%)	217 (71%)	217 (48.6%)	217 (56%)
295-296 (25 %)	295 (16%)	295 (26%)	295 (31.5%)	295 (47.4%)
405-406 (87%)	405 (63%)	405 (81%)	405 (57.4%)	405 (66.9%)
917-918 (42%)	915 (20%)	915 (32%)	917 (26.5%)	917 (46.8%)

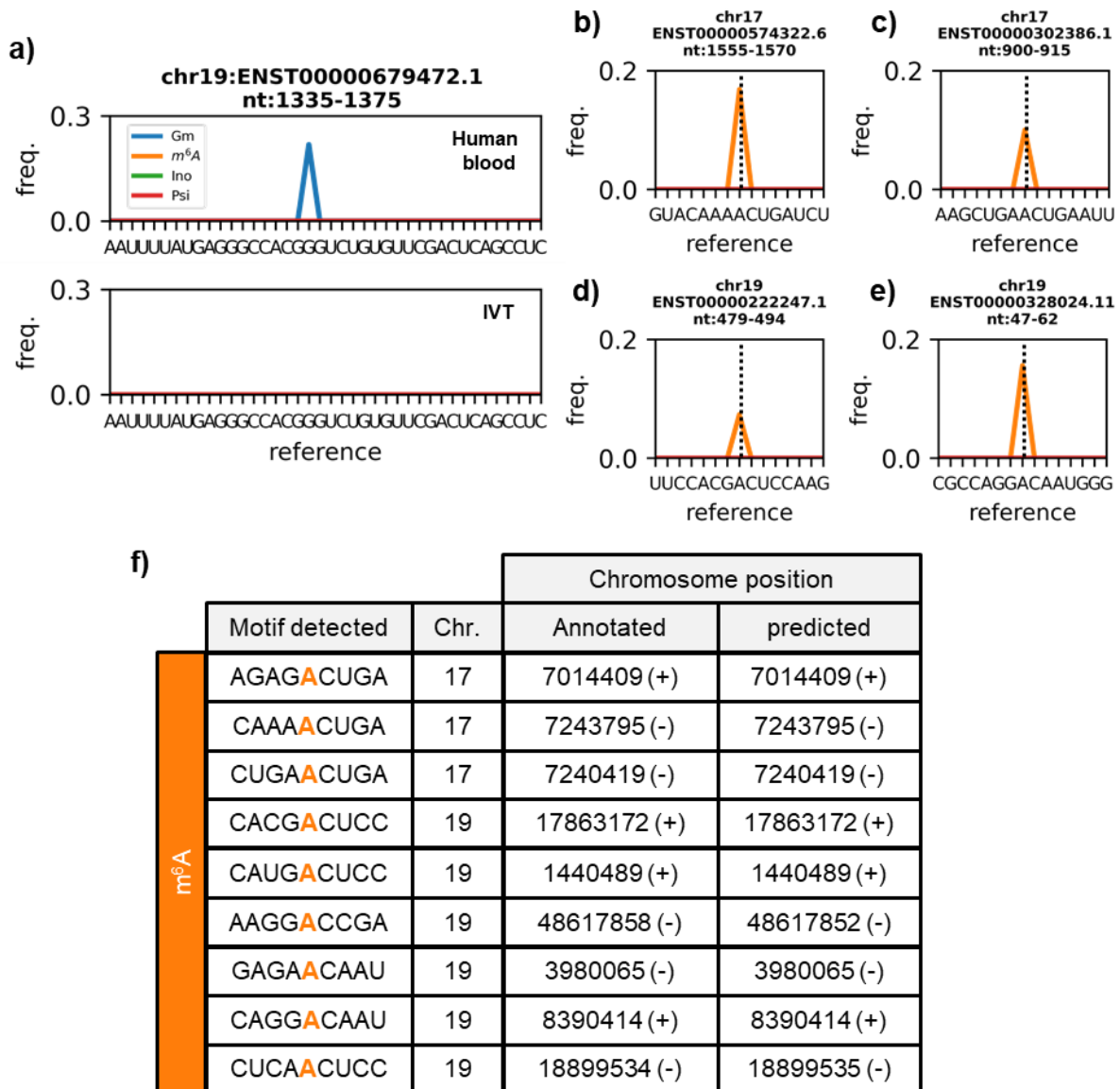
Supplementary Table 4: Comparison between literature (reference 61 in the main text), Dorado Ψ model, and ModiDeC on the Ψ sites detected in this work. The table shows both position and modification frequency. Sample 1 = Nucleus; Sample 2 = Cytoplasm

SI 6. ModiDeC analysis on human blood sample

In this section, it is possible to find additional ModiDeC analysis on human blood sample. Supplementary Fig. 8 shows additional m⁶A sites that ModiDeC was able to detect during the analysis on the gene TP53.

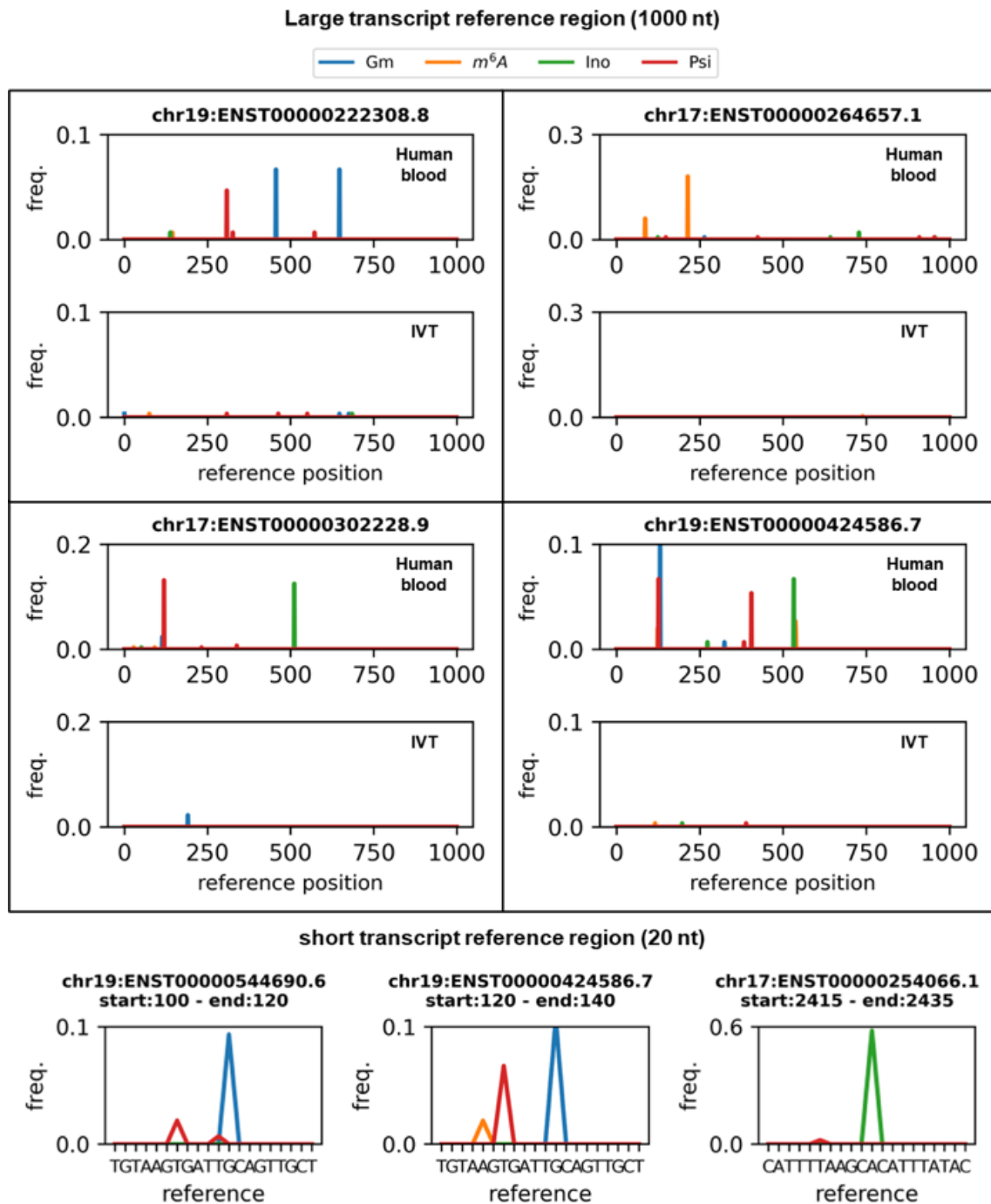


Supplementary Fig. 8: Human blood RNA analysis using ModiDeC. Additional m⁶A sites that ModiDeC was able to detect during the analysis on the gene TP53. These peaks are currently not annotated in RMBASE v3.0 database. Although further validation analyses are needed for these peak values, we found them interesting enough to show for follow-up work.



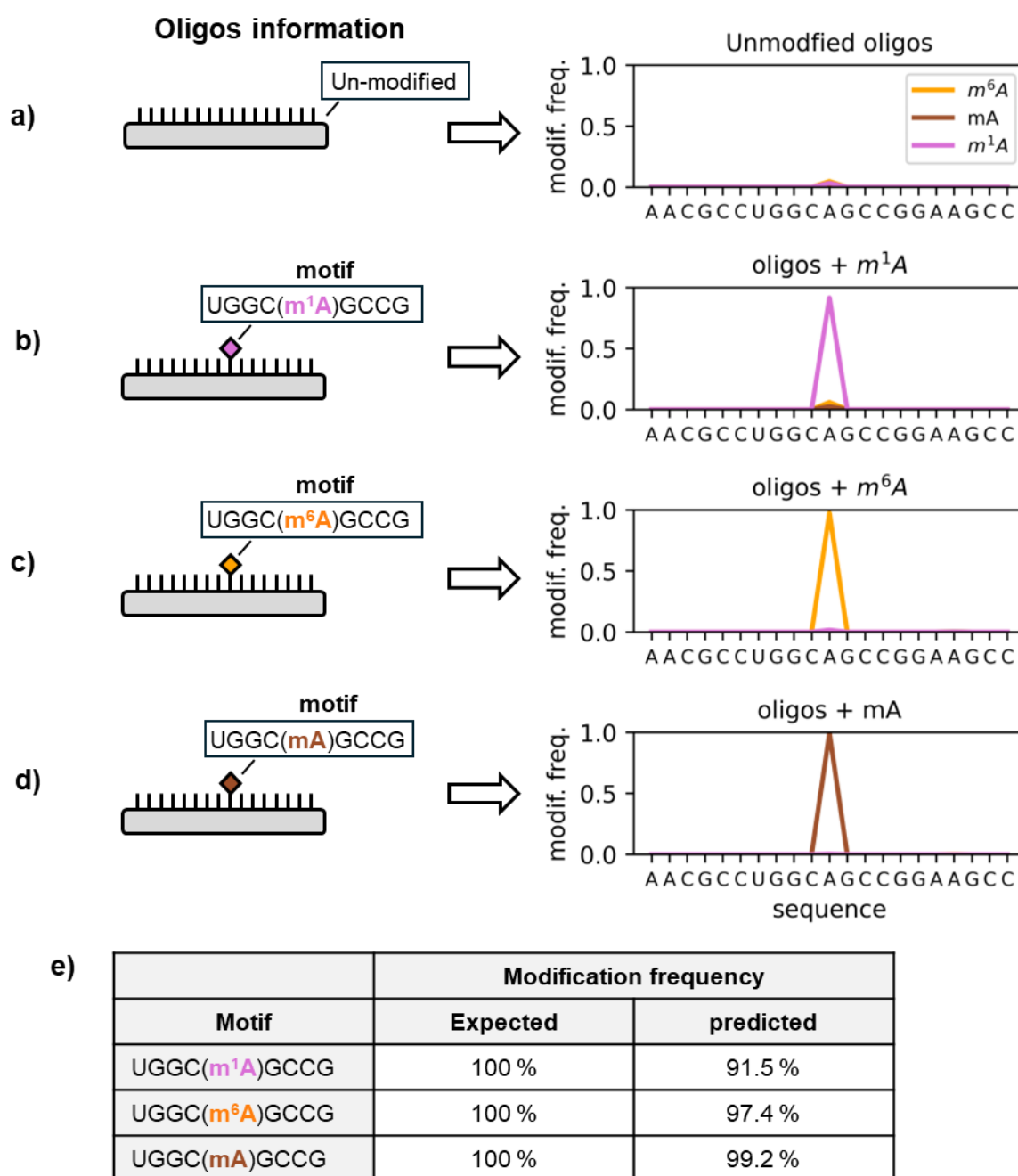
Supplementary Fig. 9: a–e) Examples of annotated m⁶A peaks (black dashed lines) from the atlas database that were detected by ModiDeC. f) Table showing the annotated m⁶A peaks from the atlas database detected by ModiDeC. The detected motif (the orange letter represents the m⁶A modification), chromosome, and annotated and predicted chromosome positions are shown.

Supplementary Fig. 9 shows additional m⁶A positions identified by ModiDeC that were also confirmed by the m⁶A-Atlas database. Supplementary Fig. 9 a-d shows four of the nine positions obtained by ModiDeC with their corresponding modification frequency. Figure 9f shows a table summarizing the ModiDeC results. These are the recognized motifs, the annotated and the predicted chromosome positions. Additional analysis and the broader large transcript reference region are shown in Supplementary Fig. 10.



Supplementary Fig. 10: ModiDeC analysis on human blood RNA. a) ModiDeC analysis and modification frequency detection on selected transcripts with a large reference region of 1000 nt. Each subplot shows the comparison between normal and IVT human data for a specific transcript. The lowest intensity peaks in the IVT data shows that peaks in normal data are not false positives. b) 20 nt zoom of the ModiDeC analysis for a few transcripts. In this case, the x axis shows the reference nucleotides, and RNA modifications are correctly assigned to the nucleotide base to which they belong.

Supplementary Fig. 11 shows the ModiDeC analysis performed on RNA oligos with same sequence context and containing different RNA modification types with similar chemical structure.



Supplementary Fig. 11: ModiDeC analysis on RNA oligos with same sequence context and containing different RNA modification types with similar chemical structure. The analysis was performed on (a) unmodified oligos, (b) m¹A, (c) m⁶A, and (d) mA modified oligos. (e) Table showing the expected and ModiDeC-predicted modification frequency for each motif.

SI 7. ModiDeC RNA modification features extraction

Here we compared the transcription motifs detected by ModiDeC, that were confirmed by the m⁶A atlas database (see also Supplementary Fig. 9f), with those used for training the neural network. The comparison is shown in Supplementary Table 4. The table shows that only one of the detected motifs exactly matches a training motif. The other detected motifs match partially or with a combination of training motifs. For example, looking the second row in the table and the motif CAAAACUGA, it can be reproduced by the CAAA and ACUGA parts of two different training motifs. Gray letters show the missing trained nucleotides in the detected motif.

	Motif detected	Trained motif
m ⁶ A	AGAGACUGA	AGAGACUGA
	CAAAACUGA	CAAAACAAU + AGAGACUGA
	CUGAACUGA	AGAGACUGA
	CACGACUCC	AGUGACUCC + CUUACUCC
	CAUGACUCC	AGUGACUCC
	AAGGACCGA	CAGGACCGA
	GAGAACAAU	GAGAACAC +/or CAAACAAU
	CAGGACAAU	CAGGACAC +/or CAAACAAU
	CUCAACUCC	CUUACUCC

Supplementary Table 5: Comparison between m⁶A-modified motifs detected in ModiDeC and trained motifs. The detected motifs are not always exact matches with the trained ones. Gray letters show the missing trained nucleotides in the detected motif. The detected motifs were also confirmed by the Atlas database (see also Fig. 5g in the main text).

SI 8. ModiDeC user graphic interface

We created three graphic user interfaces (GUIs) that can be used to generate, train, or analyze data with ModiDeC in a user-friendly environment (Fig. 6, main text). The ModiDeC GUI can be used in several ways, from retraining the neural network to directly analyzing an aligned sample using a pretrained neural network. In this section, we give an overview of how these three GUI can be used individually or all together in sequence to generate a personalized ModiDeC neural network. We decided to create the GUIs to give the opportunity to adapt and customize ModiDeC for specific problems. A full tutorial on how to use ModiDeC can be found in the GitHub page (see Code availability in the main text for the link).

Fig. 6 (main text) shows a general overview of the ModiDeC GUIs. The GUIs are titled "ModiDeC data curation", "ModiDeC training", and "ModiDeC analysis". Each section represents a different step in retraining or using ModiDeC for a final analysis. The GUI "ModiDeC data curation" (Fig. 6, main text, top) helps the user to create data sets for training the neural network. As one can see from the graphical interface, it is possible to personalize the creation of the training data to a great extent. It is important during pre-processing to correctly assign the output labels and relative signal information for proper and correct neural network training. In fact, important features such as "modification position", "type of modification" or "chunk size" (or sliding window size) can be personalized as input for the graphical interface for ensuring that signal-to-sequence information is stored correctly and learned during training. To start the "ModiDeC data curation" GUI, three elements are needed: 1) the pod5 folder (where measured pod5 files are stored); 2) the aligned .bam file of your data (stored in a folder); 3) the ONT level k -mer table for re-squiggle. It is important that data are base-called using the --emit-moves flag during base calling because it will change the .bam file for re-squiggle. After opening these three elements using the GUI, it is possible to set the variables that lets the user decide how to chop and label the signal during the re-squiggle. After selection, pressing the "start resquiggle" button will start the GUI generating the training data in a selected save folder. In detail, the GUI will pre-process the data in two phases. The first phase consists of the re-squiggle (signal-to-sequence assignment), where signal and the sequence are cut in personalized windows and stored in the memory. In the second phase, the modification is assigned correctly to the sequence and the output label is generated consecutively to have precisely assign

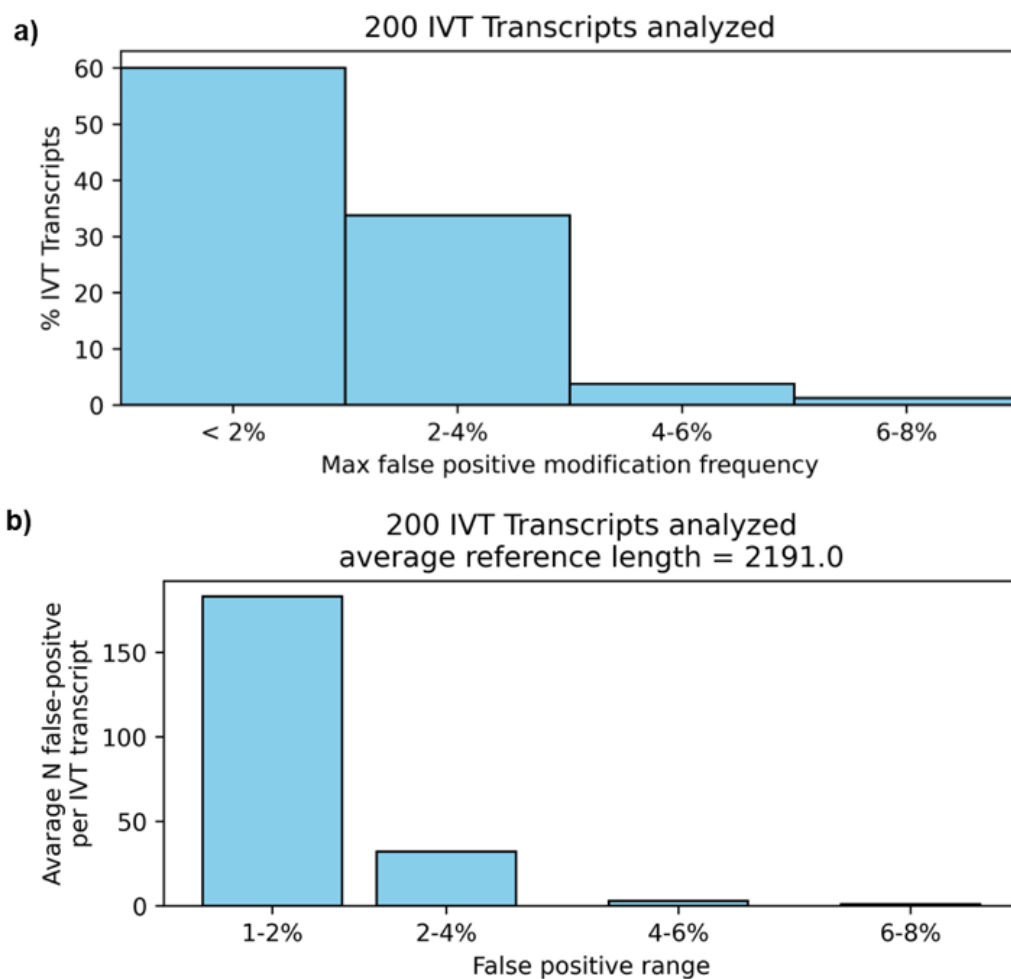
a modified sequence to a signal. Next, raw signal, sequence, and labels are stored in a .npz file and they are ready to be used as training in the “ModiDeC training” section.

In the “ModiDeC training” GUI (Fig. 6, main text, middle), it is possible to retrain the neural network using the training data created in the previous step. Loading the folder of the training data using the graphical interface and setting neural networks parameters such as epochs and batch size, it is possible to start training the neural network. After the training, the trained model is saved in a selected folder and a personalized ModiDeC for a selected problem is ready to use in the analysis section.

Having a training model, it is then possible to start the “ModiDeC analysis” GUI (Fig. 6, main text, bottom). In this user interface we can load our model and pod5 and bam files to analyze the data that we are interested in. After initialization (a step to load the model and some variables), it is possible to select the number of reads to analyze and then start the neural network analysis. After pressing the button “Start analysis with neural network”, ModiDeC analyzes the raw signal of the data and results are saved in the working folder with the name “ModiDeC_analysis.npz”. To check the analysis, clicking the “visualize results” button a new window will open, and the analysis can be visualized in an interactive way.

SI 9. False positive analysis on IVT

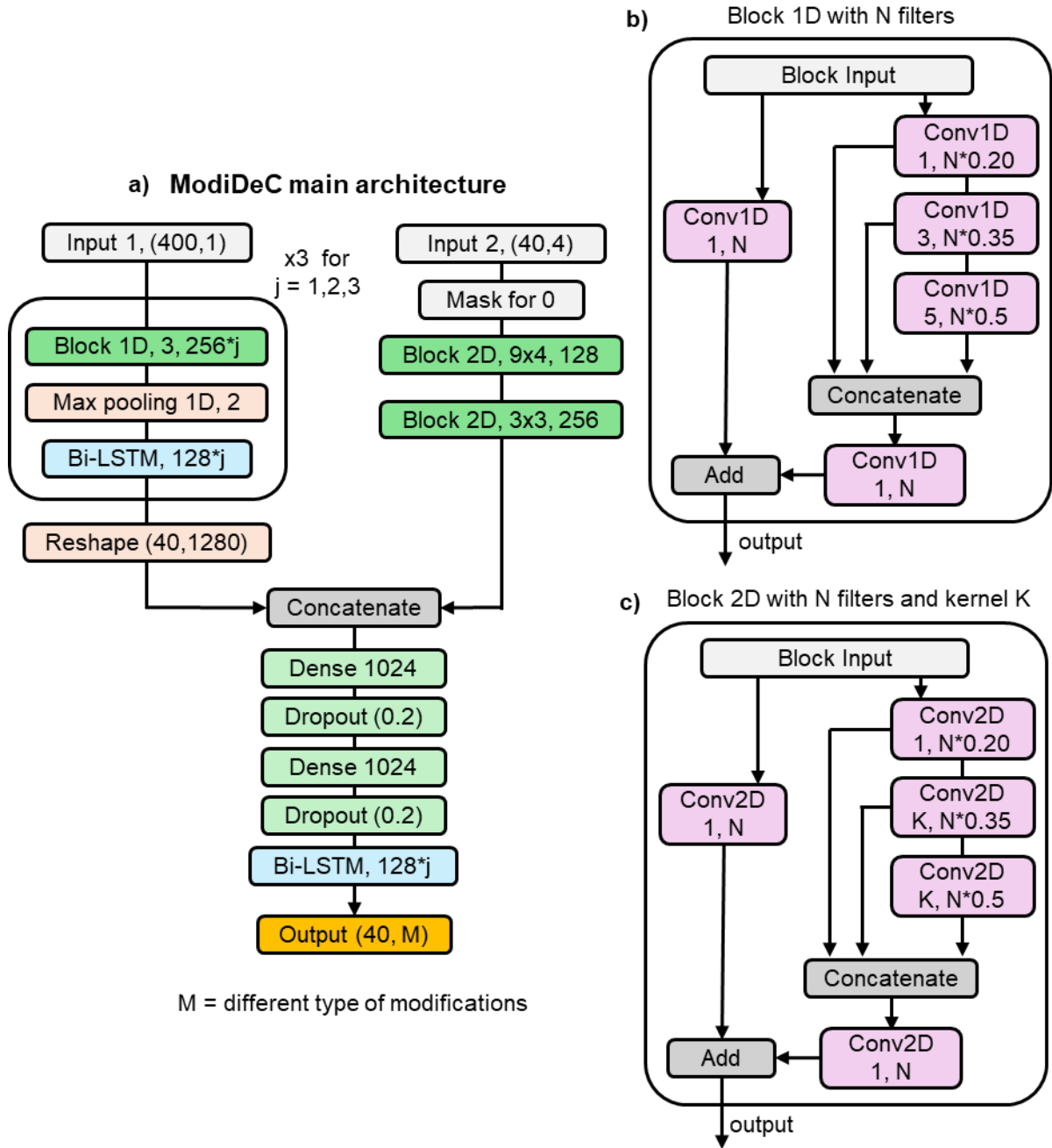
This section focuses on the analysis of false positive rates on ITV transcript. 200 transcripts were used for the analysis. In detail, we analyzed two things: i) the maximum false positive rate per transcript (Supplementary Fig. 11a); ii) the average amount of false positive detected per transcript and their value range (Supplementary Fig. 11b). The false positive rate analysis shows that ~99% of the data have a maximum false positive value below 6%, while only 1% have a false positive value between 6-8%. We also calculated the mean value $\mu = 1.9 \%$ and standard deviation $\sigma = 1.5 \%$, suggesting a ModiDeC minimum threshold $\approx 5\%$ for biological data (calculated by $2 * \sigma$ from the μ value). The analysis to quantify the number of false positives shows that total amount of false positive, on average, is about 200 for a transcript length of 2191 nucleotide. Moreover, the analysis shows that most of the false positive are between 1-4%.



Supplementary Fig. 12: a) Quantification of IVT transcripts max false positive rate over 200 transcripts. About 99% of transcripts has a maximum false positive below 6%. **b)** quantification of false positive amount per transcript and their false positive value range. The analysis was performed over 200 IVT transcripts. The average length of the transcripts is 2191 nucleotides.

SI 10. ModiDeC architecture

ModiDeC is a multi-layer neural network with ~20M of parameters that combines Z-score normalized raw signal, one-hot encoding sequences and new personalized Inception blocks to classify multiple RNA modifications at the same time from ONT measurements. Supplementary Fig. 11a shows the ModiDeC architecture used for this work. The neural network has two inputs: i) the raw signal input, or input 1), that has a dimension of (400,1); ii) the one-hot sequence input, or input 2, that has a dimension of (40,4). The two inputs are processed differently compared to each other. The input



Supplementary Fig. 13: **a)** ModiDeC main architecture. It is a two-input neural network that combine raw signal and one-hot encoded sequence to detect RNA modification on the sequence. **b)** Personalized inception block used for the Input 1 pathway. **c)** Personalized inception block used for the input 2 pathway.

1 goes through convolution 1D (Block 1D, Supplementary Fig. 11b), Bi-LSTM and maxpooling layers. After three times, the output is reshaped into (40,1280) to match the first dimension of Input 2 for Concatenation. In parallel, Input 2 is also processed but only with convolution 2D layers (Block 1D, Supplementary Fig. 11c). In the end, the two processed inputs are concatenated and sent through Dense, Bi-LSTM layers. The Output last layer is a Dense layer with a sigmoid activation function with a

dimension (40, M), where M is the number of different types of modifications to analyze. The Sigmoid activation function is used for the last layer to better predict output with different length size.